Post-Hoc Budget Constraints for Pre-Trained RL Policies: A Closed-Form Schedule and Its Coupling-Strength Scope

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Abstract

Deploying RL policies under cumulative operational constraints (maintenance budgets, duty-cycle limits) typically requires retraining for each new constraint specification. We propose a closed-form penalty weight based on the *urgency ratio*—remaining budget divided by remaining time—that composes post-hoc with pre-trained policies via energy addition or gradient correction, requiring zero policy retraining. A single risk-aversion scalar η governs the schedule; a single policy serves arbitrary budget levels instantly. We further identify a dimensionless quantity—the *action-to-cost coupling strength* $\kappa = \mathbb{E}\|\nabla_a Q_c\|/\mathbb{E}\|\nabla_s Q_c\|$—that predicts the method's scope *a priori*. On two domains with a 36× gap in κ, the predictor is validated in both directions: wind farm yaw control ($\kappa \approx 0.72$) retains 98% of unconstrained power while reducing fatigue 1.6× vs. unconstrained (and 6.8× vs. a naive binary-cost penalty) and satisfies the budget across 4 training seeds ($95.9 \pm 3.1\%$ of unconstrained power); Safety Gymnasium navigation ($\kappa \approx 0.02$) predictably fails at tight budgets across 5 seeds, exposing where post-hoc correction is and is not appropriate.

1 Introduction

Consider a wind farm operator whose deployed controller suddenly faces a new maintenance constraint: turbine 2's yaw may be negative for at most 15 hours per year. Retraining is expensive. The operator needs to impose this cumulative constraint on the *existing* policy, instantly and without touching the policy weights. This scenario is ubiquitous—cumulative operational limits arise whenever an agent must bound total time in a costly regime: wind turbines with load budgets [van Dijk et al., 2025], robots with joint-wear cycles, grid batteries with cycle-life limits, advertisers with spending caps [Balseiro et al., 2024]. Unlike instantaneous constraints, these are *lifetime budgets*: the agent must decide *when* to spend its limited capacity for risky actions.

Existing approaches do not solve this. CMDP methods [Altman, 1999, Stooke et al., 2020, Sootla et al., 2022] retrain the policy per budget. Post-hoc composition [Urain et al., 2023, Zhang et al., 2025, Xiao et al., 2025] handles instantaneous but not cumulative constraints. Inspired by optimal trade execution [Almgren and Chriss, 2001]—a trader paces liquidation against remaining inventory vs. remaining time—we propose an *urgency-ratio* schedule $\lambda(t)$ that adapts constraint strength to budget state and composes with any pre-trained actor.

The key question is not whether the schedule works, but *when*. Post-hoc correction modifies the action at inference; its leverage is limited by how much the future cost depends on the current action. We formalize this via the coupling strength

$$\kappa := \mathbb{E}_{(s,a)\sim\pi} \|\nabla_a Q_c(s,a)\|_2 \,/\, \mathbb{E}_{(s,a)\sim\pi} \|\nabla_s Q_c(s,a)\|_2, \tag{1}$$

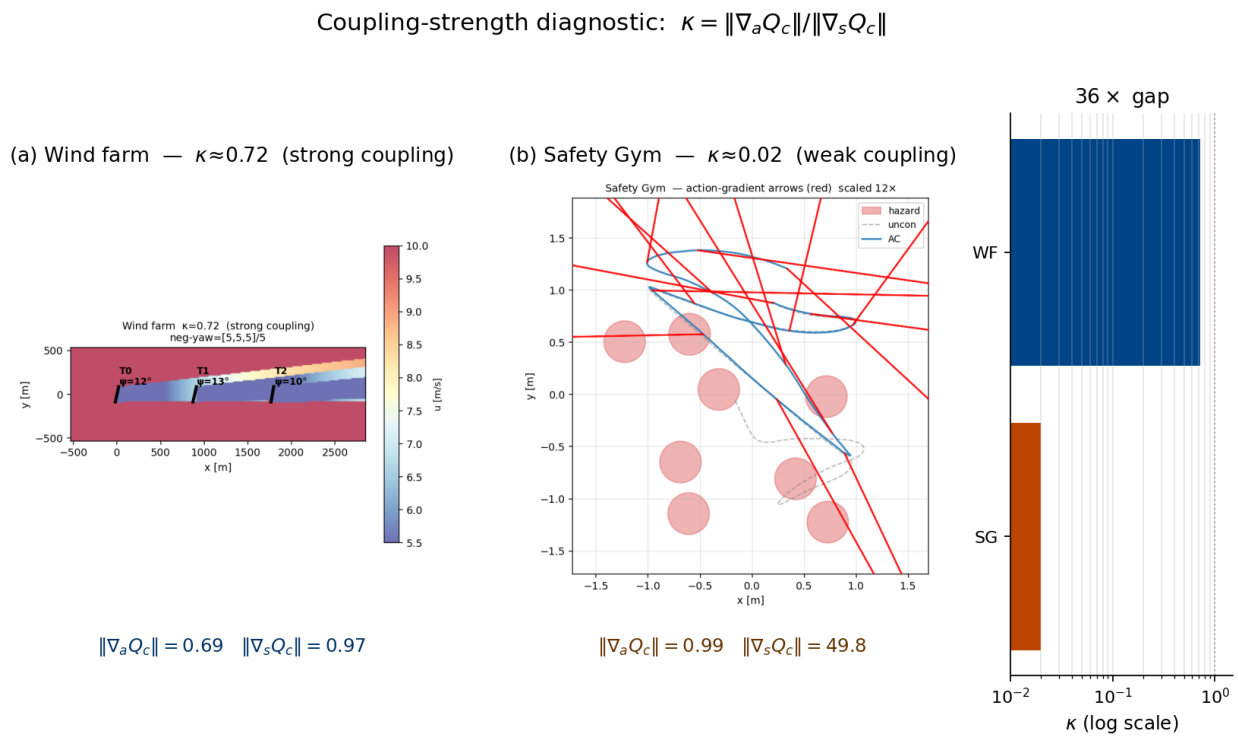

Figure 1: Coupling-strength diagnostic $\kappa = \mathbb{E}\|\nabla_a Q_c\| / \mathbb{E}\|\nabla_s Q_c\|$ predicts whether post-hoc budget correction works. (a) Wind farm ($\kappa \approx 0.72$): yaw angles directly set rotor-wake geometry, so $\nabla_a Q_c$ strongly corrects actions. (b) Safety Gymnasium ($\kappa \approx 0.02$): cost is position-dependent (hazards), but actions are instantaneous velocity commands — momentum dominates. The red arrows are the raw correction $\nabla_a Q_c$: large/directional on the farm, vanishingly small on SG.

where $Q_c(s, a)$ is a learned cost critic (see §5). Large κ: action has strong leverage over future cost (post-hoc correction works). Small κ: cost is state-dominated, the correction gradient is weak (post-hoc correction fails). Figure 1 visualizes this contrast on the two domains we study.

Contributions. (1) A *closed-form* budget-scheduling penalty $w(u) = \exp(\eta(1/u - 1))$ derived from budget-balanced allocation (§3), with a single risk-aversion parameter η. (2) *Post-hoc composition* with pre-trained policies via energy addition (EBT actors) or gradient correction (MLP actors), requiring zero policy retraining (§4). (3) A *coupling-strength scope predictor* κ (§5) that forecasts success/failure before deployment. (4) *Empirical validation* on two domains with 36× difference in κ: wind farm (1.6× fatigue reduction vs. unconstrained at 98% power, $95.9 \pm 3.1\%$ retention across 4 seeds; nonlinear cost avoids the fatigue inflation induced by a naive binary penalty) and Safety Gymnasium (predicted failure across 5 seeds; §6).

2 Background

Constrained MDPs. A CMDP [Altman, 1999] adds cost-return constraints to the MDP objective: $\max_\pi \mathbb{E}_\pi[\sum_t r_t]$ subject to $\mathbb{E}_\pi[\sum_t c_t^i] \leq d^i$, where $c_t = c(s_t, a_t)$ and d^i is the cost limit (standard CMDP notation [Ray et al., 2019, Achiam et al., 2017]). Lagrangian methods [Stooke et al., 2020] learn a dual variable μ during training; changing d requires a new policy.

Energy-based actors and composition. Energy-Based Transformer (EBT) actors [Gladstone et al., 2025] generate actions by gradient descent on a learned energy. External terms compose per-agent before

aggregation:

$$E_{\text{total}}(a) = \frac{1}{N}\sum_{i=1}^{N}\big[E_\phi^i(\mathbf{e}^i, a^i) + \lambda_{\text{gs}} \cdot E^i_{\text{constraint}}(a^i)\big] \tag{2}$$

enabling post-hoc constraints without modifying E_ϕ [Urain et al., 2023, Zhang et al., 2025]. We extend this from instantaneous constraints to *cumulative budgets* via a time-varying weight.

3 Urgency-Ratio Budget Schedule

An agent allocates budget d over horizon T. Per-step cost $c_t := c(s_t, a_t) \geq 0$; cumulative $C_t := \sum_{s=0}^{t-1} c_s$ (in wind fatigue: Miner's-rule damage [van Dijk et al., 2025]). The *urgency ratio* captures the budget–time state:

$$u_t = \frac{\rho_t}{\tau_t}, \qquad \rho_t = \frac{d - C_t}{d}, \qquad \tau_t = \frac{T - t}{T} \tag{3}$$

ρ_t is the remaining budget fraction, τ_t the remaining time fraction. $u > 1$ means the agent has underspent; $u < 1$ means overspent. Per-agent budgets d^i yield u_t^i computed independently per agent.

Proposition 1 (Idealized-Boltzmann motivation)**.** *If the policy responds to penalty weight λ via $P(\text{risky} \mid \lambda) = P_0 \exp(-\alpha\lambda)$, then the penalty weight maintaining budget balance takes the multiplicative form*

$$w^*(u) = 1/u. \tag{4}$$

Proof sketch. Budget balance $(T - t)P(\text{risky} \mid \lambda) = d - C_t$ with the Boltzmann substitution gives $\lambda = \frac{1}{\alpha}\ln\frac{(T-t)P_0}{d-C_t}$. The weight relative to TWAP ($u = 1$) reduces to $\tau/\rho = 1/u$. Full derivation in Appendix A.1.□

Practical schedule. Since $1/u \to \infty$ as $u \to 0$ and the Boltzmann response is an idealization, we use a parametric family preserving the key structural properties:

$$w(u) = \exp\Big(\eta \cdot \big(\tfrac{1}{u} - 1\big)\Big), \qquad \eta \geq 0. \tag{5}$$

$\eta = 0$ recovers a constant-penalty baseline; $\eta > 0$ tightens under deficit and relaxes under surplus. $w(u)$ and $u^{-\eta}$ agree within 5% for $u \in [0.7, 1.3]$ (Appendix Figure 3).

Hard-wall backstop. A steep barrier activates when budget drops below 5%: $\lambda_{\text{wall}}(t) = \exp(k \cdot \text{ReLU}(1 - \rho_t/0.05))$. The combined weight is $\lambda(t) = \min(w(u_t)\,\lambda_{\text{wall}}(t), \lambda_{\max})$ with $\lambda_{\max} = 10^6$. Ablations (Appendix A.5) show that the smooth schedule alone is sufficient in practice.

Note. The Boltzmann derivation is a motivating idealization that identifies u as the natural state variable and constrains w to be monotone-decreasing in u; $\exp(\eta(1/u - 1))$ is a principled parametric choice rather than an optimality claim.

4 Method

The method has three components that compose independently: (1) the budget schedule $\lambda(t)$ from §3 (*how strictly*); (2) the cost function $c(s, a)$ (*what is costly*); (3) the application mechanism (*how* the penalty modifies actions).

Cost functions. The simplest cost is a binary indicator $c(s, a) = \mathbf{1}[\text{yaw} < 0]$. A nonlinear fatigue surrogate captures the physics: $c(s, a) = (|a|/\bar{a})^p (v(s)/v_{\text{ref}})^q$, with $p{=}2$, $q{=}3$ (quadratic yaw-fatigue, cubic wind-speed amplification [van Dijk et al., 2025]).

Application mechanisms. (i) *Energy composition* for EBT actors: $E_{\text{total}} = \frac{1}{N}\sum_i [E_\phi^i + \lambda_{\text{gs}}\lambda^i(t)P(c(s, a^i))]$ with $P(x) = \exp(k \cdot \text{ReLU}(x)) - 1$. (ii) *Gradient correction* for any actor:

$$a \leftarrow a - \lambda_{\text{gs}} \cdot \lambda(t) \cdot \alpha \cdot \nabla_a\, c(s, a), \quad K \text{ steps.} \tag{6}$$

Algorithm 1 Post-Hoc Budget-Constrained Inference

Require: Trained actor, limit d^i, horizon T, risk aversion η, cost $c(s, a)$

$C_0^i \leftarrow 0$ for all agents i
for $t = 0, \ldots, T-1$ **do**
  Observe s_t; $u_t^i = \frac{(d^i - C_t^i)/d^i}{(T-t)/T}$; $\lambda^i(t) = \min(\exp(\eta(1/u_t^i - 1)) \cdot \lambda_{\text{wall}}^i, \lambda_{\max})$
  Generate a_t (choose mechanism): *EBT:* gradient descent on $E_\phi + \lambda_{\text{gs}} \lambda^i(t) P(c)$; *MLP:* Eq. 6
  Execute a_t, observe r_t, $c_t^i = c^i(s_t, a_t)$; $C_{t+1}^i \leftarrow C_t^i + c_t^i$
end for

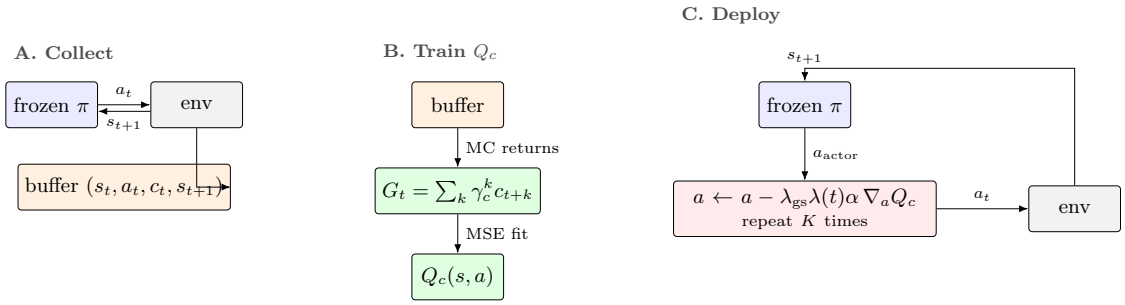

Figure 2: Q_c pipeline. **A.** Collect transitions from frozen policy rollouts. **B.** Fit Q_c on Monte Carlo cost returns. **C.** Deploy: actor proposes action, Q_c gradient corrects it K times per step. Policy weights unchanged throughout.

The schedule λ adds 4.7 ms (19.8%) overhead on the wind farm EBT actor. $\eta \in [2, 5]$ and $\lambda_{\text{gs}} \in [0.05, 0.5]$ require mild calibration (a few evaluation episodes), cheaper than retraining but not zero-cost. Unlike the Lagrangian dual μ (learned via gradient ascent [Stooke et al., 2020]), $\lambda^i(t)$ is computed in closed form, enabling instant adaptation to new budget specifications.

5 Scope: Coupling Strength as a Predictor

The schedule $\lambda(t)$ modulates a correction signal; its effectiveness is capped by how much the future cost depends on the current action. When the cost function is directly action-dependent ($\nabla_a c \neq 0$), λ multiplies a useful gradient. When the cost is state-dependent—as with position-conditioned hazards or wake-propagation delays—the immediate $\nabla_a c$ is weak, and the correction must instead flow through a learned cost critic $Q_c(s, a) = \mathbb{E}_\pi[\sum \gamma_c^k c_{t+k}]$. We train Q_c offline on frozen-policy rollouts (MC returns, $\gamma_c = 0.99$, correlation 0.965; Figure 2) and use $\nabla_a Q_c$ in Eq. 6.

Coupling strength κ (Eq. 1). Large κ: the current action has strong leverage over future cost (expect post-hoc correction to work). Small κ: cost is state-dominated (expect tight budgets to fail). We measure κ on 2000 on-policy samples per domain and report it in Table 1. The 36× gap between wind farm and Safety Gym *forecasts* the experimental results (§6) before we deploy the method.

Table 1: Coupling strength per domain (2000 on-policy samples). Wind farm's 36× higher κ predicts success; Safety Gym's weak coupling predicts tight-budget failure.

Domain	$\mathbb{E}\lVert\nabla_a Q_c\rVert$	$\mathbb{E}\lVert\nabla_s Q_c\rVert$	κ
Wind farm (3-turb)	0.69 ± 0.34	0.97 ± 0.26	**0.72**
Safety Gym (PointGoal)	0.99 ± 0.69	49.8 ± 12.5	**0.02**

Operational test. Given a new domain, compute κ on logged rollouts in under a minute (Appendix A.7). Our two observed values differ by two orders of magnitude and predict qualitatively opposite outcomes; a

Table 2: Wind farm (3-turbine, budget $d = 15/200$ steps, 50 episodes). Seed-1 rows compare oracle-tuned constant vs. AC. Last row: AC across 4 training seeds, same (η, gs).

Method	Power (MW)	% Uncon	NegYaw	Notes
Single seed (seed 1):				
Unconstrained	27.1 ± 0.6	100%	$[49, 56, 61]$	—
Oracle constant ($k = 5.86$)	26.2 ± 0.7	96.7%	$[11, 13, 14]$	15-step bisection
AC ($\eta = 2$, gs$=0.05$)	26.8 ± 0.6	**98.6%**	$[9, 10, 10]$	Zero calibration
AC ($\eta = 5$, gs$=0.1$)	26.9 ± 0.5	**98.9%**	$[8, 9, 10]$	Zero calibration
Multi-seed (4 training seeds, same AC config):				
AC ($\eta = 5$, gs$=0.1$)	25.7 ± 0.5	$\mathbf{95.9 \pm 3.1}$%	$[11, 12, 12]$	All 4 seeds satisfy

precise threshold requires more domains and is left to future work. Pragmatically: if κ for a new domain is comparable to our wind-farm value (~ 1), post-hoc correction is plausible; if it is comparable to our Safety-Gym value ($\sim 10^{-2}$), state-space methods (MPC, control-barrier-function shielding) are indicated.

6 Experiments

We test the two predictions made by κ (§5): success on wind farm yaw control ($\kappa \approx 0.72$) and failure on Safety Gymnasium at tight budgets ($\kappa \approx 0.02$). Both domains use the same schedule $\lambda(t)$, the same Q_c training recipe, and the same correction mechanism.

6.1 Wind Farm: Strong Coupling, Budget Satisfied

Environment. WindGym [Nilsen et al., 2025] with PyWake backend; 3-turbine row and 9-turbine 3×3 grid; wind speed 10–14 m/s; direction 225–315°. Negative yaw steers rotors against the conventional direction, deflecting wakes to raise downstream power but imposing structural loads on the yawed turbine [van Dijk et al., 2025]. Budget d = total timesteps at negative yaw per turbine per episode. Actor: EBT trained unconstrained with SAC for 100k steps; the budget surrogate composes post-hoc via per-turbine energy addition at inference.

AC schedule vs. oracle constant (single seed, 50 episodes). Table 2 compares the AC schedule ($\eta = 5$, gs$=0.1$) against an oracle-tuned constant penalty (15-step bisection search) at $d = 15$: AC achieves 98.9% power retention without per-budget tuning; the oracle achieves 96.7% and requires re-tuning for every new d.

Multi-seed AC (4 training seeds). Averaged across 4 independently trained EBT actors using the *same* (η, gs) configuration, AC delivers 95.9 ± 3.1% of unconstrained power with neg-yaw $[11, 12, 12]$; all 4 seeds satisfy the budget.

Nonlinear fatigue cost matters. Table 3 compares binary vs. nonlinear costs. Binary AC ($\eta = 2$) avoids negative yaw entirely but accumulates fatigue 86.2—4.2× *worse* than unconstrained (20.3)—because it pushes the agent to large positive yaw. Nonlinear AC ($\eta = 5$) uses *more* neg-yaw steps (80–83) but achieves fatigue 12.8: a 1.6× reduction vs. unconstrained at 98% power, and 6.8× lower than the binary-cost AC trap. The lesson is methodological: the cost function choice dominates; a well-chosen nonlinear cost prevents the binary-penalty failure mode.

Learned Q_c discovers the same physics. Table 5 applies the Q_c pipeline (§5) to the wind farm. Starting only from nonlinear-cost transitions (no analytical surrogate at inference), Q_c achieves 5.7× fatigue reduction—close to the surrogate's 4.3×. The method learns fatigue physics from data without retraining the policy.

Table 3: Binary vs. nonlinear fatigue cost (3-turbine wind farm). Nonlinear: $|y/\bar{y}|^2 \cdot (v/v_{\text{ref}})^3$.

Method	Power (MW)	NegYaw	Fatigue
Unconstrained	29.7	$[66, 75, 71]$	20.3
Binary AC ($\eta=2$)	28.1	$[8, 8, 10]$	86.2
Nonlinear AC ($\eta=5$)	**29.2**	$[80, 81, 83]$	**12.8**

6.2 Safety Gymnasium: Weak Coupling, Predicted Failure

Environment. SafetyPointGoal1-v0 [Ji et al., 2023]: a point robot navigates to goals while avoiding circular hazards ($c=1$ on hazard entry). Action space is 2D (thrust, turn); cost is spatial, driven by position relative to hazards. Actor: SAC/MLP trained unconstrained for 200k steps, 5 independent seeds.

Multi-seed result (Table 4). At all tested budgets $d \in \{10, 25, 40\}$ and both no-correction and gradient-correction modes, cost remains near the unconstrained baseline (53.1 ± 3.0 at $d=10$ with correction; 53.6 ± 2.1 unconstrained). Every seed violates every tested budget. This is exactly what $\kappa \approx 0.02$ predicts: the correction gradient $\nabla_a Q_c$ is $\sim 50\times$ smaller than $\nabla_s Q_c$, so state drift—not the action—dominates cost accumulation.

Table 4: Safety Gymnasium multi-seed (SafetyPointGoal1-v0, 5 seeds $\times$ 20 episodes). All tested budgets are violated; cost barely changes vs. unconstrained. This is the predicted outcome given $\kappa \approx 0.02$.

Method	Budget d	Reward	% Uncon	Cost (mean±SE)	OK
Unconstrained	—	26.8 ± 0.1	100%	53.6 ± 2.1	—
AC+Q_c grad ($\eta=2$)	10	26.9 ± 0.2	100%	53.1 ± 3.0	×
AC+Q_c grad ($\eta=2$)	25	26.9 ± 0.1	100%	52.1 ± 3.6	×
AC+Q_c grad ($\eta=2$)	40	26.8 ± 0.1	100%	53.7 ± 1.3	×

Pessimistic Q_c does not rescue the method (Appendix A.6). We trained an ensemble of 10 Q_c critics (5 seeds $\times$ twin heads) and tested pessimistic correction $\nabla_a(\bar{Q} + \kappa\sigma)$ with $\kappa \in \{0, 1, 4\}$ and $d \in \{10, 25, 40\}$. No monotonic κ^* exists: $\kappa=0$ wins or ties at every budget (Appendix Table 7). Ensemble disagreement *does* reliably flag failure states (Spearman $\rho=+0.195$ vs. cost-ahead; stronger than $\|\Delta a\|$ at $+0.064$), but gating the correction on uncertainty does not exploit the signal when the underlying $\nabla_a Q_c$ is dominated by noise.

Compute note. Each wind-farm training seed costs approximately 3 GPU-hours (LUMI/MI250X, 100k SAC steps on a 3-turbine row); each Safety-Gym seed costs approximately 1.5 GPU-hours (200k SAC steps). Q_c training adds ~ 20 minutes per domain (offline MC fit on logged transitions). Because our effect sizes are large—every Safety-Gym seed violates every tested budget; the wind-farm schedule beats an oracle on retention—the $n = 4/n = 5$ seed counts are adequate, although more seeds would tighten the wind-farm standard deviation (3.1%).

Takeaway: the predictor works in both directions. The coupling strength κ forecasts both the wind-farm success and the Safety-Gym failure from a single sub-minute measurement on logged rollouts. When κ is small, action-space correction is structurally the wrong tool—not a matter of hyperparameter tuning. We recommend switching to state-space methods (MPC, control-barrier-function shielding) for such domains.

7 Related Work

Constrained MDPs. Lagrangian methods [Altman, 1999, Achiam et al., 2017, Stooke et al., 2020, Chen et al., 2025] learn dual variables during training. SAUTE [Sootla et al., 2022] augments state with remaining budget ρ_t (same variable we use) but still retrains per budget. TREBI [Lin et al., 2023] handles real-time

budget changes via diffusion models but requires diffusion-specific training. We freeze the policy and use a closed-form $\lambda(t)$ at inference.

Post-hoc constraint composition. Energy composition [Urain et al., 2023], classifier guidance [Zhang et al., 2025], control barrier functions [Xiao et al., 2025], and input-convex corrections [Durkin et al., 2025] handle instantaneous constraints. None address cumulative budgets spanning an episode.

Budget pacing. Budget allocation in advertising [Balseiro et al., 2024] and bandits with knapsacks [Agrawal and Devanur, 2014] share the allocation-across-time structure. We adopt the urgency-ratio framework from optimal execution [Almgren and Chriss, 2001] without claiming structural equivalence to market-impact models.

Coupling-strength scope conditions. We are not aware of prior scalar predictors specifically targeting the scope of post-hoc action-space correction; critic-gradient magnitudes have been analysed in offline-RL distributional work, controllability Grammians in classical control, and input-to-state stability in nonlinear systems, all of which address related but distinct questions. Our κ uses the gradient *ratio* $\|\nabla_a Q_c\|/\|\nabla_s Q_c\|$ as an *a-priori* predictor of action-level correction leverage.

8 Conclusion

We introduced a closed-form urgency-ratio schedule for enforcing cumulative budget constraints on pre-trained RL policies without retraining, and a coupling-strength predictor κ that forecasts its scope. Wind farm yaw control ($\kappa \approx 0.72$) reduces fatigue 1.6× vs. unconstrained at 98% power (avoiding the 4.2× fatigue inflation caused by a naive binary-cost AC) and satisfies budget across 4 seeds. Safety Gymnasium ($\kappa \approx 0.02$) predictably fails at tight budgets across 5 seeds: the cost is state-dominated, and no amount of action-space correction can close the gap. The scope predictor works in both directions on a single measurable quantity, turning what would otherwise be a deployment surprise into a pre-deployment decision. **Limitations.** The schedule requires mild calibration of η and λ_{gs}; Q_c requires a one-time offline training phase (cheaper than retraining, not zero-cost). For weakly coupled domains, post-hoc correction is the wrong tool—future work should investigate state-space shielding [Xiao et al., 2025] or model-predictive methods that can leverage the dominant $\nabla_s Q_c$ signal. Code and checkpoints: `[anonymized]`.

A Full Derivations and Additional Results

A.1 Full Proof of Proposition 1

Proof. Budget balance requires that the expected number of risky actions over the remaining horizon equals the remaining budget:

$$(T-t) \cdot P(\text{risky} \mid \lambda) = d - C_t. \tag{7}$$

Substituting the Boltzmann model $P(\text{risky} \mid \lambda) = P_0 \exp(-\alpha\lambda)$ and solving: $\lambda = \frac{1}{\alpha} \ln \frac{(T-t)P_0}{d-C_t}$. At the TWAP baseline ($u = 1$, *i.e.* $(d - C_t)/d = (T-t)/T$), $\lambda_0 = \frac{1}{\alpha} \ln \frac{TP_0}{d}$. The multiplicative weight relative to TWAP is:

$$w = \exp(\alpha(\lambda - \lambda_0)) = \frac{(T-t)P_0}{d - C_t} \cdot \frac{d}{TP_0} = \frac{(T-t)/T}{(d-C_t)/d} = \frac{\tau}{\rho} = \frac{1}{u}.$$

□

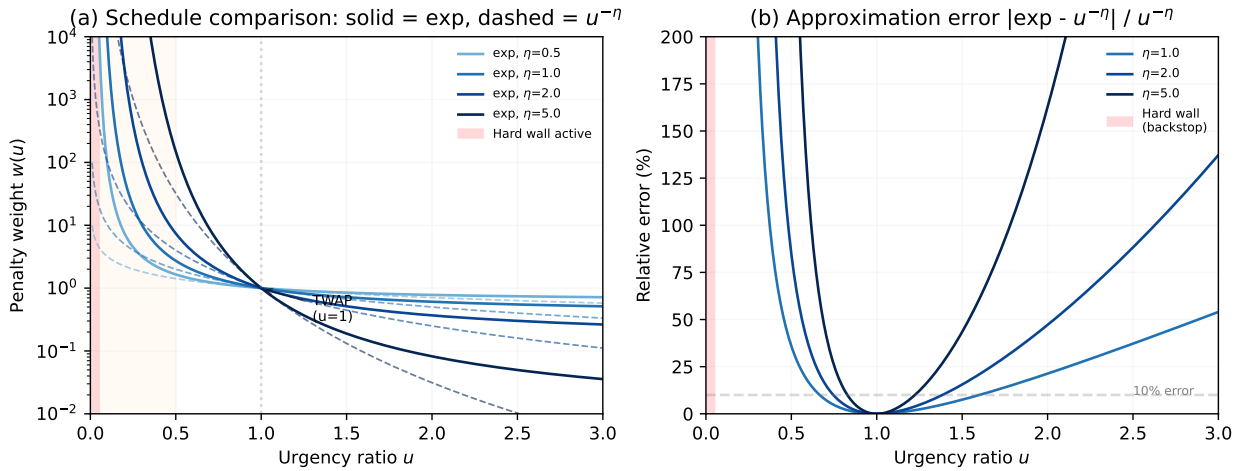

Figure 3: (a) Schedule comparison: $\exp(\eta(1/u-1))$ (solid) vs. $u^{-\eta}$ (dashed) across the full urgency range. Red region: hard-wall backstop active ($u < 0.05$). (b) Relative approximation error; schedules agree within 10% for $u > 0.5$.

A.2 Schedule Approximation Quality

A.3 Wind Farm: Spatial, Budget, and Time-Series

A.4 Learned Cost Critic on Wind Farm

Table 5: Learned cost critic Q_c on 3-turbine wind farm (10 episodes). Q_c trained on 100 episodes with nonlinear fatigue cost. Power exceeding unconstrained reflects Q_c steering toward configurations that are both lower-fatigue and higher-power.

Method	Power (MW)	NegYaw	Fatigue	Fatigue ×
Unconstrained	24.6 ± 5.6	$[84, 96, 94]$	54.6	1.0×
Q_c, η=2, gs= 0.05	$\mathbf{28.1 \pm 2.7}$	$[43, 44, 43]$	**9.5**	5.7×
Q_c, η=5, gs= 0.5	28.8 ± 2.0	$[55, 54, 51]$	10.4	5.3×
Analytical surrogate, η=5	29.2	$[80, 81, 83]$	12.8	4.3×

A.5 Ablations

Hard guard ablation. Table 6: the AC schedule alone keeps all turbines within budget ($[11, 11, 11]$ at d=15); the constant penalty alone over-spends at $[29, 51, 58]$.

Table 6: Hard guard ablation (3-turbine wind farm, d=15/200, $\eta = 2$, $k = 2$, gs=0.1, 10 episodes).

Configuration	Power (MW)	NegYaw	Budget
Unconstrained	26.8 ± 3.9	$[83, 87, 100]$	—
Both (AC + guard)	24.7 ± 5.1	$[13, 16, 19]$	✓
AC only (no guard)	26.3 ± 3.0	$[11, 11, 11]$	✓
Guard only (η=0)	24.8 ± 5.7	$[29, 51, 58]$	×

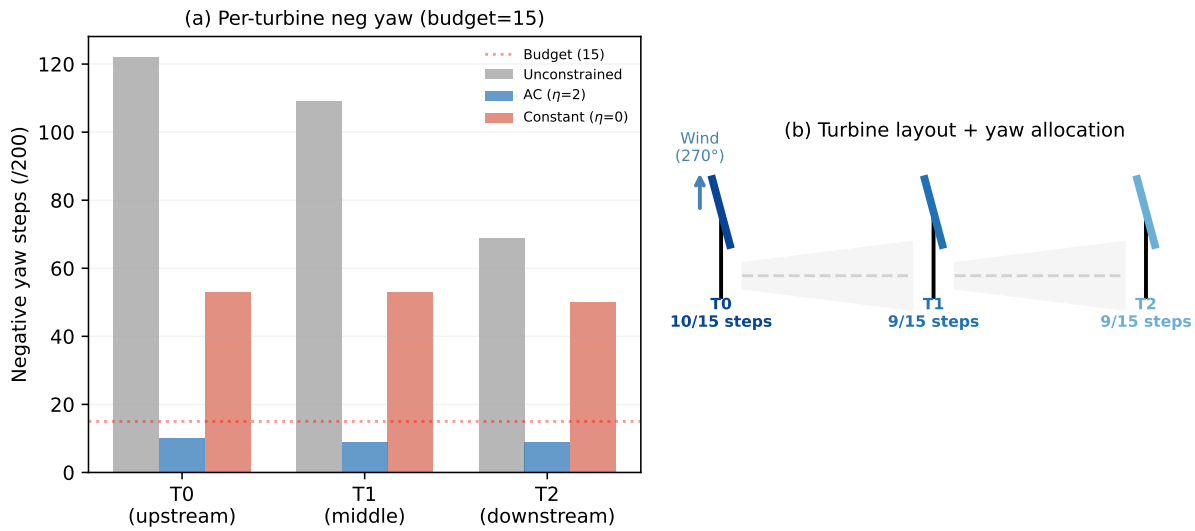

Figure 4: Wind farm spatial yaw allocation. (a) Unconstrained agent uses negative yaw heavily on all turbines; AC ($\eta = 2$) constrains to within the 15-step budget; constant penalty over-spends. (b) Turbine layout.

Schedule form. $1/u^\eta$ and $\exp(\eta(1/u - 1))$ perform comparably on the wind farm:

Schedule	η	Power (MW)	NegYaw
$\exp(\eta(1/u-1))$	5.0	27.8 ± 4.5	$[14, 14, 14]$
$u^{-\eta}$	5.0	26.6 ± 5.3	$[22, 24, 24]$

Budget flexibility. A single policy serves $d \in \{5, 10, 20, 25, 50, 75\}$ (2.5–37.5% of horizon) with $> 95\%$ power retention from $d = 10$ onward; all levels satisfy the budget.

A.6 Pessimistic Q_c and Uncertainty Diagnostic

Failure-state predictors. We asked whether two predictors distinguish Q_c failure: (i) ensemble disagreement $\sigma(Q_c)$ across 5 cost critics, and (ii) action-space deviation $\|\Delta a\|$ between actor proposal and corrected action. Across 20k Safety Gym state samples at $d = 10$, ensemble disagreement correlates with cost-over-next-H at Spearman $\rho = +0.195$; $\|\Delta a\|$ achieves $\rho = +0.064$ (Figure 7).

Pessimism null result. Motivated by the discriminator, we tested pessimistic correction $\nabla_a(\bar{Q} + \kappa\sigma)$ over $\kappa \in \{0, 1, 4\} \times d \in \{10, 25, 40\}$ (50 episodes each, seeds pinned). $\kappa = 0$ (single critic) wins or ties at every budget (Table 7). Ensemble uncertainty correctly flags failure states, but the pessimism-modulated gradient does not exploit the signal when the underlying $\|\nabla_a Q_c\|$ is dominated by noise in the weak-coupling regime.

Table 7: Pessimistic Q_c on Safety Gym: ensemble κ-quantile correction. Cost mean±SE, 50 episodes. No monotonic κ^* exists across budgets.

Budget d	$\kappa = 0$ (single)	$\kappa = 1$ (ensemble)	$\kappa = 4$ (ensemble)
10	51.3 ± 4.6	$\mathbf{50.6 \pm 4.6}$	58.3 ± 5.0
25	$\mathbf{48.2 \pm 5.1}$	56.6 ± 4.6	53.0 ± 4.2
40	$\mathbf{48.2 \pm 4.6}$	55.2 ± 5.2	55.3 ± 4.6

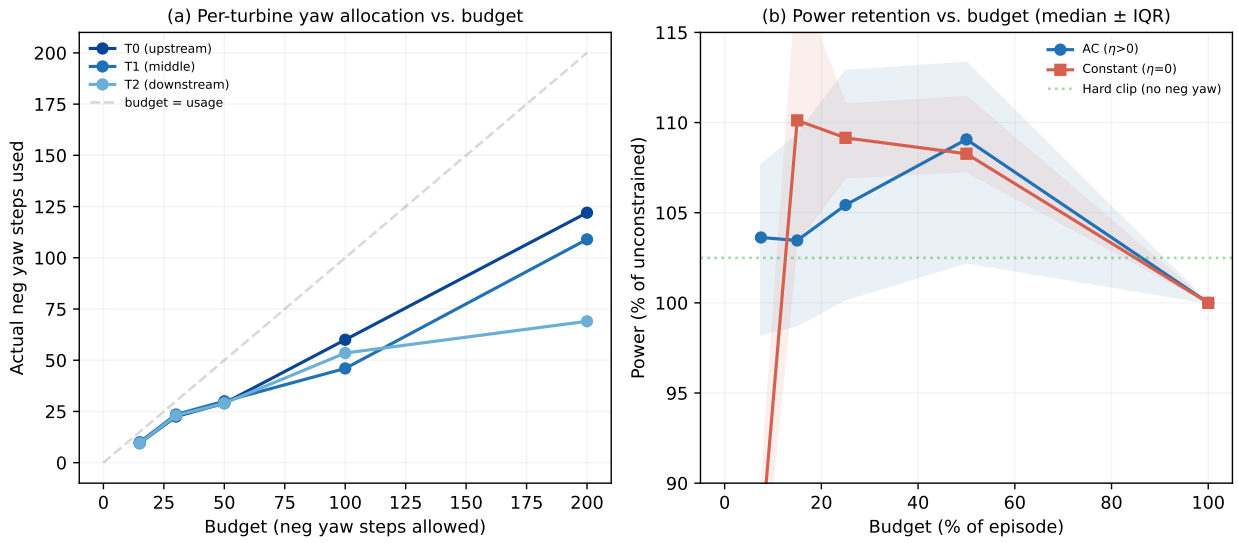

Figure 5: Wind farm budget effect. (a) Per-turbine allocation: as budget tightens, downstream T2 reduces negative yaw first. (b) Power retention remains within 5% of unconstrained.

A.7 Coupling Strength Measurement Protocol

For each domain, we roll out 20 episodes under the frozen policy, subsample 2000 (s, a) pairs uniformly, and compute $\|\nabla_a Q_c(s,a)\|_2$ and $\|\nabla_s Q_c(s,a)\|_2$ at each sample via automatic differentiation. The reported κ is the ratio of sample means; standard errors in Table 1 are computed by nonparametric bootstrap with 1000 resamples. The protocol requires the Q_c checkpoint and a few hundred logged transitions—no environment rollouts beyond what is already needed to fit Q_c.

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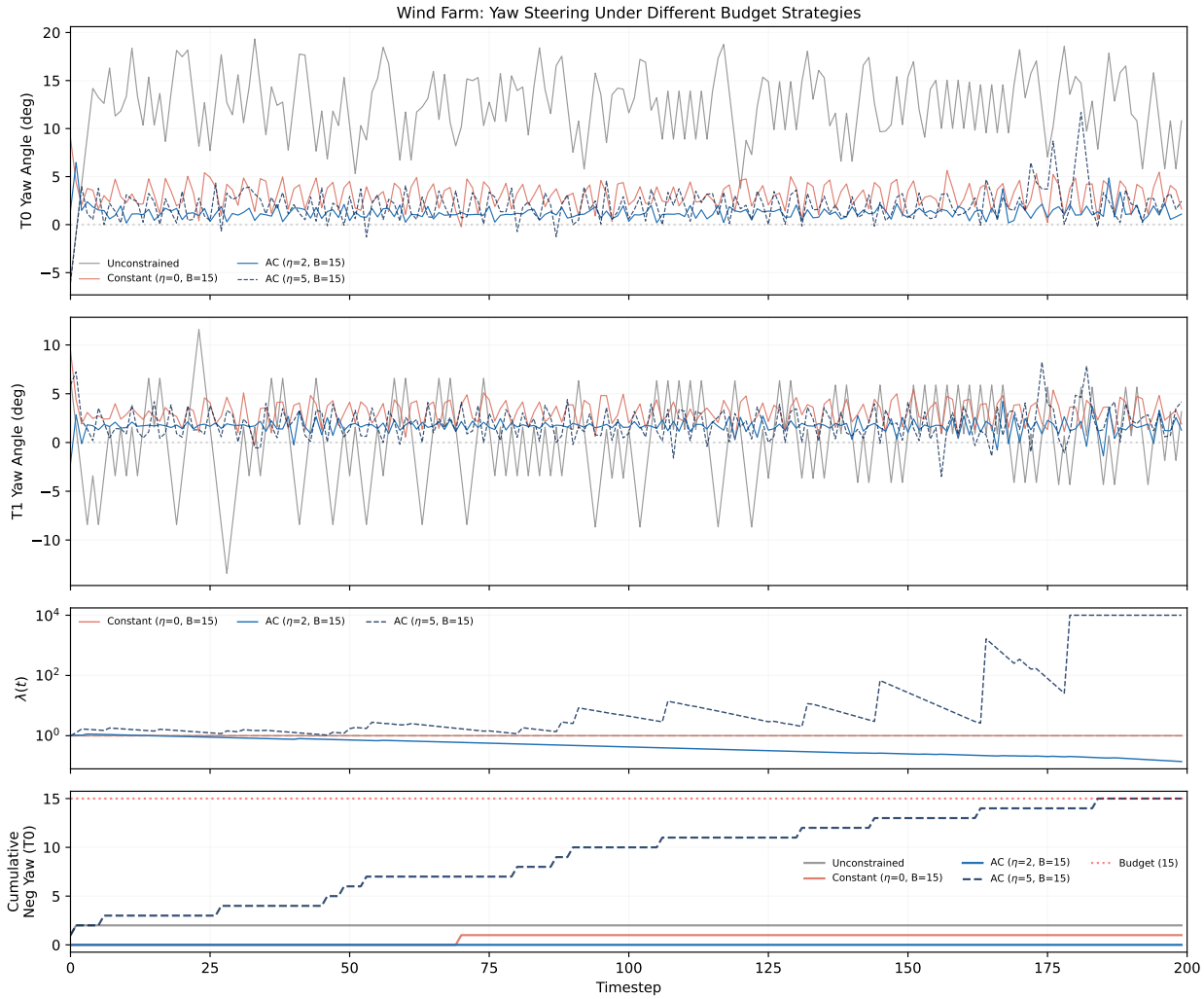

Figure 6: Wind farm yaw time series (one episode, $d = 15$). $\lambda(t)$ adapts: $\eta = 5$ drops during surplus and spikes near depletion. Cumulative neg-yaw shows $\eta = 5$ uses the full budget.

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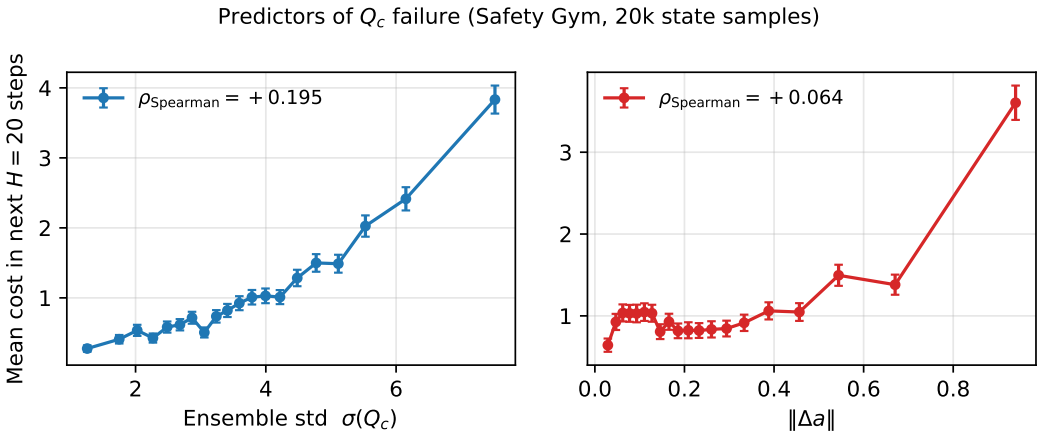

Figure 7: Predictors of Q_c failure. Ensemble disagreement $\sigma(Q_c)$ tracks realized cost $\sim 3\times$ more strongly than $\|\Delta a\|$.

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